INCH-POUND

MIL-PRF-23714D
10 March 2022
SUPERSEDING
MIL-K-23714C
04 September 1989

PERFORMANCE SPECIFICATION

KITS, SAMPLING, FOR AVIATION FUELS, REUSABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers detailed requirements for a reusable fuel sampling kit and a fuel sampling replacement kit.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to Commanding Officer, Naval Air Warfare Center Aircraft Division Lakehurst, System Standardization and PHS&T Branch, Code BL32600, Mail Stop 120-3, Route 547, Joint Base MDL, NJ 08733-5100 or emailed to LKE-NAVAIRDocComments@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database at https://assist.dla.mil

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-6054	Drum, Metal - Shipping and Storage
MIL-DTL-19834	Plates, Identification or Instruction, Metal Foil, Adhesive Backed, General Specification for
MIL-PRF-24635	Coating System, Weather-Resistant, Exterior Use
MIL-PRF-26514	Polyurethane Foam, Rigid or Flexible, for Packaging

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-129	Military Marking for Shipment and Storage
MIL-STD-3010	Test Procedures for Packaging Materials and Containers
MS27684	Drums, Metal - Shipping and Storage, 3 to 12 Gallons

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN NATIONAL STANDARDS INSTITUE (ANSI)
AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4	Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available at www.ansi.org or www.asq.org.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE) INTERNATIONAL

SAE AMS-STD-595	Colors Used in Government Procurement

(Copies of this document are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.2.

3.2 Kits. The kits furnished under this specification shall contain the items listed in Table I (see 6.2).

3.2.1 Procurement. A fuel sampling kit shall be provisioned when it is ordered and not substituted with a fuel sampling replacement kit. The components of the fuel sampling kit ensure that samples:

a. Arrive intact.
b. Provide proper secondary hazardous material containment.

TABLE I. Kit components.

FUEL SAMPLING KIT	FUEL SAMPLING REPLACEMENT KIT
Four sampling bottles with closures	Four sampling bottles with closures
Cushioning material	Four sampling tags
Container	
Tamperproof seal	
Four sampling tags	

3.2.2 Components.

3.2.2.1 Sampling bottles. The sampling bottle shall be 32 ±0.375 ounce capacity made of borosilicate or United States Pharmacopeia (USP) Type III soda-lime glass (see 6.6). The bottle shall be cylindrical with an outer diameter of 3.75 ±0.125 inches and an overall height of 8.25 ±0.125 inches. The weight of the bottle shall be 16 ±0.625 ounces (see figure 1). The closure shall be made of black molded phenol-formaldehyde and lined with a polyethylene conical liner. The sampling bottles shall be one of the two following colors:

a. Colorless: Specifically for Navy use.
    1. NSN 8125-00-378-9994
b. Amber.
    1. NSN 6640-01-688-0898

3.2.2.2 Cushioning material. The cushioning material shall be built up or a molded shape composed of elastic polyether urethane foam conforming to Type I, Class 2, Grade C of MIL-PRF-26514. The density of the cushioning material shall be 3 ±0.5 pounds per cubic foot.

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No filler material shall be included. The built-up cushion half shall be comprised of no more than three pieces cemented together: one section for the main body of the bottle, one section for the neck, and one section a solid cushion pad. The cushioning shall be designed to form a snug fit when placed in the exterior shipping container and loaded with the four sampling bottles (see figure 1). The cushioning shall consist of two separate parts for easy sampling container accessibility. The cushioning shall provide sampling container immobility and protection from impact and vibration during handling and shipment.

3.2.2.3 Exterior container. The container shall be a metal drum conforming to MIL-DTL-6054 and MS27684-7 (see figure 1). The interior and exterior of the container shall be coated with a red semi-gloss enamel conforming to MIL-PRF-24635, in lieu of the olive draft specified in MIL-DTL-6054. The red color shall conform to No. 21136 of SAE AMS-STD-595. A primer shall be used under the red enamel.

3.2.2.3.1 Marking. The exterior container of the fuel sampling kit shall be marked in accordance with MIL-STD-129 and in one inch high letters as follows:

AVIATION FUEL SAMPLES
REUSABLE CONTAINER
DO NOT DESTROY

The fuel sampling replacement kit shall be marked for shipment in accordance with MIL-STD-129. The part number is 80132-P001-4 and the National Stock Number is 8115-00-717-8572.

3.2.2.3.2 Nameplates. Nameplates conforming to Type I of MIL-DTL-19834 shall be attached to the upper portion of the drum, one on the exterior and one on the interior. The names shall be marked as follows:

Sampling Container Fuel Kit
Part No. 80132-P001
NSN 8115-00-719-4111
Cushioning
Part No. 80139-P001-1
NSN 8115-00-719-4825

3.2.2.4 Sampling tag. The sampling tag shall be of white cloth, size 8, 6.25 inches by 3.125 inches (NSN 8135-00-178-9193). Tags shall be strung with not less than 12 inches of polished cotton twine. Tags shall be furnished with the printed markings as shown on figure 2. Durable black ink shall be used for the markings. Type size shall be as follows:

Aviation test sample	16 point type
Classification, Routine, Special	12 point type
All other markings	10 point type

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3.3 Assembled fuel sampling kit. The kit shall be furnished completely assembled and with the drum lid loosely sealed. The closures shall be fitted on the bottles to keep them clean, dry, and ready for use. The tags with the string shall be included but not attached to the sampling bottles. The four bottles shall be positioned upright with one in each cushioning cavity. The upper cushioning pad shall be secured in place, and the exterior container loosely sealed to the extent necessary to prevent loss.

3.3.1 Closure of container. The container shall be sealed by placing the gasket and the lid in position on the container. For the bolt-type lid, effect closure by inserting bolt through lugs on closure rings. Tighten nut and bolt to 6 ±1 foot pound torque pressure. On the twist lock container, effect closure by inserting locking tool in each of four locks at the top of the container and exerting a twisting motion.

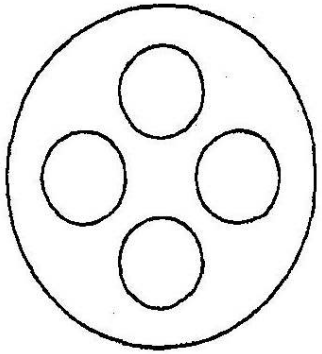
3.3.1.1 Tamperproof seal. The closure shall be capable of accommodating a crimped seal through the hole in the closure bolt, through closure ring lugs, or at twist lock to effect a tamperproof seal. The crimped seal is not necessary when the container is empty (see 6.2).

3.4 Assembled fuel sampling replacement kit. The replacement kit shall consist of 4 bottles and 4 tags. The closures shall be fitted on the bottles, and the tags, with strings, shall be included with, but not attached to the sampling bottles.

3.5 Impact resistance. The kits and their cushioned contents shall show no evidence of simulated fuel leakage. The drums shall show no deformation which will affect reclosing and resealing when tested as specified in 4.4.2.

3.6 Workmanship. The kit assembly components shall be free of any defects which may affect durability, strength and serviceability.

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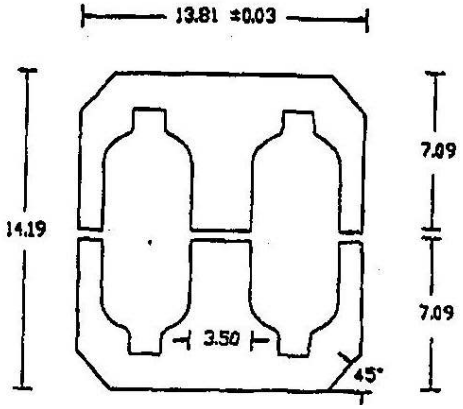


Arrangement of four
sample bottles in kit



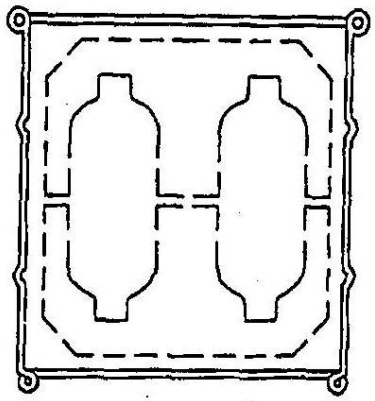
Sample bottle
32 ±0.375 ounce capacity

NOTES:
1. Dimensions are in inches.
2. Tolerances are ±0.125 inches unless otherwise specified.
3. Drawing not to scale.



Cushioning, MIL-PRF-26514

Type 1, Class 2, Grade C



Exterior Container

MIL-DTL-6054, MS27684

FIGURE 1. Kit assemblies.

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FIGURE 2. Printed sampling tag, NSN 8135-00-178-9193.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall include all the examinations and tests of this specification see (4.4.1 and 4.4.2), except the inspection of packaging (see 4.4.3). Unless otherwise specified, no material shall be submitted for acceptance under any contract or order until the first article samples prescribed in 4.2.1 have been subjected to first article inspection and pronounced satisfactory by the procuring activity, or until equipment certification has been approved by the procuring activity (see 4.2.2). However, approval of the first article samples or equivalent certification and data shall not relieve the supplier of his obligation to meet the quality conformance inspection. Fuel sampling kits shall be prepared for shipment completely assembled in accordance with 3.3 and 3.4.

4.2.1 First article samples. Unless otherwise specified in the contract or purchase order, the sample for first article inspection shall consist of one fuel sampling kit and one fuel sampling replacement kit (see 6.2 and 6.3).

4.2.2 First article waiver. If a contractor has previously furnished the kits in accordance with the requirements of this specification and his product has been found satisfactory, the requirement for a first article sample and its submittal for any subsequent contract or order may be waived at the discretion of the procuring activity (see 6.2 and 6.3).

4.3 Conformance inspection. Conformance inspection shall consist of the examinations listed under 4.4.1 and 4.4.3, and, at the discretion of the contracting officer, of the free fall drop test of 4.4.2 (see 6.4).

4.3.1 Sampling for conformance. For lot sampling purposes, a lot shall consist of all fuel sampling kits of fuel sampling replacement kits subjected to inspection at one time.

4.3.2 Selection of sample. The selection of a sample for examination or tests shall be in accordance with ANSI/ASQ Z1.4. The unit for determining lot size and the sample unit shall be as specified in the applicable examination or test paragraph.

4.4 Methods of inspection.

4.4.1 Examination of end item. Noncompliance with any specified requirement or presence of one or more defects shall constitute cause for rejection. The sample unit and unit for determining the lot size for this examination shall be one fuel kit and one replacement kit.

4.4.1.1 Examination for defects in construction, material, design, and workmanship. The fuel sampling kit shall be examined for compliance with the requirements specified in 3.2, 3.3,

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3.4, and 3.6 and for defects as listed in table II. The fuel sampling replacement kit shall be examined for compliance with the requirements and for defects applicable to the sampling bottles, closures and tags.

TABLE II. Examination of end item.

EXAMINATION	DEFECTS
Materials	Any component or material not as specified.
Construction and design	Any component missing or malformed
Sampling bottle	Not cylindrical Bottom of bottle does not rest evenly upon flat, smooth, level surface Seams or edges not smooth Presence of blisters over 3/32 inch, stones over 1/32 inch and any cracks, chips, dirt or foreign matter embedded in glass Cap does not provide tight fitting closure
Cushioning material	Not uniformly molded or built-up to contain bottles Evidence of filler material in composition Not clean
Exterior container	Finish not as specified, gaskets missing or damaged, covers do not fit properly, sharp edges and metal splinters, presence of dirt, rust or foreign matter.
Sampling tag	Not cloth Not strung with polished cotton twine, 12 inches or longer in length Printed markings incorrect, illegible, omitted
Assembly	Assembled kits not as specified (see 3.3, 3.3.1, 3.3.1.1 and 3.4)
Workmanship	Any defect in construction or appearance such as improper chime of drum, bottle closure not uniformly lined

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4.4.1.2 Examination for dimensional defects. The fuel sampling kit and replacement kit shall be examined for dimensional defects. Any dimension which does not meet the requirement within the tolerance specified shall be scored as a defect.

4.4.2 Impact resistance test. The fuel sampling kit shall be marked in quarters top and bottom and tested in accordance with MIL-PRF-24635, Method 5007, Free Fall Drop Test, procedure C for cylindrical containers using level A drop height (see 3.5). At the discretion of the contracting officer, these tests shall be repeated on each production lot of kits.

4.4.3 Packaging inspection. The preservation, packing and marking shall be inspected to verify conformance to section 5. The sample unit and unit for determining lot size for this examination shall be one palletized load.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The sampling kits covered by this specification are intended to be used for transporting samples of aviation fuel conforming to ASTM D910 or MIL-DTL-5624 by surface or air. The kit is designed to be durable and reusable except for the bottles and tags. The replacement kit contains replacements for the disposable bottles and tags.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Quantity.

c. When first article is required (see 3.1).

d. Whether fuel sampling kit or fuel sampling replacement kit is required (see 3.2).

e. Whether impact resistance test (see 3.5) is required (see 4.4.2).

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f. Whether sealing wire and uncrimped metal seal is furnished with drum (see 3.1.1.1).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.2. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Sampling bottles. Bottles manufactured of amber, borosilicate or USP Type III soda-lime glass are required to protect the integrity of the fuel sample. The amber color prevents photochemical reactions and the borosilicate or soda-lime glass resists pH change. Bottles manufactured of colorless, borosilicate or USP Type III soda-lime glass are for visual sampling of free water and/or sediment/particulates in the fuel.

6.5 Subject term (key word) listing.

Fuels
Fuel testing
Sampling bottles

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - GL
Navy - AS
Air Force - 69

Review activities:
Army - AV, SM
Navy - MC
Air Force - 99

Preparing activity:
Navy – AS
(Project 8115-2019-001)

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